

Homework 3 — Functions (the refactor)

Homework 3 — Functions: the refactor

Due Sunday 4 October, 23:59, on Canvas — the same Sunday as Homework 2, on purpose.

This one is short, and it is about *shape*, not new problems. Take the two programs you wrote for Homework 2 and rewrite them the way every program is written from now on: **functions that return, and one main() that reads and prints**. Then add one new function.

What you submit

One file, A_Matricula_hw3.py, in the exam format (name comment on line 1, algorithm in """), containing:

1. **is_leap_year(year)** — returns True or False. No input(), no print() inside.
2. **descending(a, b, c)** — returns the three numbers ordered from largest to smallest. (Return three values separated by commas: return x, y, z. Python hands them back together; in main you catch them as x, y, z = descending(a, b, c).)
3. **classify_triangle(a, b, c)** — the new one — returns "not a triangle", "equilateral", "isosceles" or "scalene". Three sides make a triangle only when all are positive and each pair adds up to *strictly more* than the third.
4. **main()** that asks the user which of the three to run (1, 2 or 3), reads the values it needs, calls the function, and prints the result. A last line that calls main().

Plus, as a photo or scan: **the algorithm of classify_triangle** on paper, and **your shell tests** — the calls you typed and what came back.

Shell tests — required

Run the file, then type these in the Thonny shell and write down what you got:

```
is_leap_year(2000)      True
is_leap_year(2100)      False
descending(-9, 4, 1)    (4, 1, -9)
descending(5, 5, 2)     (5, 5, 2)
classify_triangle(3, 3, 3) 'equilateral'
classify_triangle(3, 3, 5) 'isosceles'
classify_triangle(3, 4, 5) 'scalene'
classify_triangle(1, 2, 3) 'not a triangle'
```

If a function needs input() to be tested, it is built wrong — that is the whole lesson.

Rubric — published

Three functions return the right values on the shell tests (no print, no input inside)	50 %
main() reads, calls and prints; the file ends with main()	20 %
classify_triangle algorithm on paper, boundary handled (1, 2, 3 is not a triangle)	20 %
Shell tests written down as run	10 %

Two points off for each missing technical requirement (name comment, algorithm in "", file name).